

CYBERMIND Idea Description

Smart India Hackathon 2026 | Problem Statement SIH26153 | CYBERMIND

Proposed Solution

CYBERMIND is an offline, analyst-centred predictive cyber-defence platform that moves beyond the conventional detect-and-alert model. It represents an enterprise network as a time-ordered sequence of host communication graphs, learns how that state evolves, and forecasts the risk and coarse attack stage of upcoming graph windows. This gives defenders an evidence-linked view of what may happen next while an intrusion is still developing.

An analyst imports a PCAP, PCAPNG, CAP, or network-flow CSV. CYBERMIND parses the input locally, preserves file provenance, converts records into a canonical flow schema, and builds overlapping temporal graph windows in which hosts are nodes and communications are edges. A pinned graph-temporal model combines a GATv2 graph encoder, temporal Transformer, learned latent dynamics, and multi-head prediction layers to produce a short-horizon risk trajectory, coarse attack-stage estimates, uncertainty cues, and model metadata.

The forecast is integrated into a practical investigation workflow. The Command Centre connects network topology, observed flows, timelines, and forecasts. Parallel Futures applies permitted in-memory changes - such as host isolation, port blocking, or rate limiting - to copies of the current graph and compares their predicted future risk with the no-action baseline. These are decision-support simulations; the platform does not silently enforce controls on a production network.

A bounded offline assistant answers read-only questions about the loaded case. The local validation engine freezes the current forecast and records actual sandbox HTTP responses and SHA-256 response hashes. The Reports module creates an incident brief and encrypts it locally using AES-256-GCM with a fresh random key and nonce. The encrypted JSON excludes the key, allowing the file and key to be shared separately and decrypted inside CYBERMIND.

Core Innovation

- Forecasting instead of isolated classification: the model predicts aggregate future graph-window risk rather than assigning unsupported labels to every individual flow.
- Counterfactual decision support: analysts can compare the model's no-action future with simulated defensive alternatives before approving consequential action.
- Evidence separation: observations, rule-based flow triage, model forecasts, validation responses, and analyst decisions remain explicitly distinguishable.
- Offline operation: Windows desktop and Linux Docker variants keep the core pipeline independent of cloud APIs and support protected local reporting.

Expected Impact

CYBERMIND changes the defender workflow from retrospective alert review to evidence-linked prediction, simulation, and informed planning. It helps an analyst understand the present network state, anticipate a plausible near-term escalation, test reversible response options, and preserve the investigation as a protected incident brief. The result is earlier situational awareness without presenting uncertain model outputs as verified truth or autonomous enforcement.